Expand Requirement

Prompt

Act as a Business Analyst and Software Architect. Expand the initial requirement into detailed requirements for an Inventory Management System (IMS), covering:

- Core features and capabilities
- User roles and permissions
- Key workflows and business rules
- Industry-standard non-functional requirements (NFRs), including security, performance, scalability, availability, and reliability

State key assumptions and identify missing requirements that need clarification.

Response

Inventory Management System – Detailed Functional Requirements

1. Introduction

1.1 Purpose

The purpose of this system is to manage inventory operations efficiently by tracking products, stock levels, inventory movements, suppliers, and transactions across warehouses or storage locations.

The system should:

- Maintain accurate inventory records

- Reduce stock discrepancies
- Support stock inward/outward operations
- Provide real-time inventory visibility
- Generate operational and analytical reports

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2. System Scope

The Inventory Management System (IMS) will support:

- Product management
- Inventory tracking
- Stock adjustments
- Purchase receiving
- Sales dispatch
- Supplier management
- User and role management
- Reporting and analytics
- Notifications and alerts

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3. User Roles

3.1 Administrator

Responsibilities

- Configure system settings
- Manage users and permissions
- Manage master data
- Access all reports
- Audit system activities

Permissions

- Full access to all modules
- Create/Edit/Delete users
- Configure inventory rules

3.2 Inventory Manager

Responsibilities

- Monitor stock levels
- Approve stock adjustments
- Generate reports
- Manage inventory operations

Permissions

- View/manage products
- Approve stock transactions
- Access inventory reports

3.3 Warehouse Staff

Responsibilities

- Receive stock
- Dispatch stock
- Update inventory records

Permissions

- Create stock entries
- Update stock movement
- View assigned warehouse inventory

3.4 Procurement Officer

Responsibilities

- Manage suppliers
- Create purchase orders
- Track incoming inventory

Permissions

- Manage suppliers
- Create and update purchase orders

3.5 Auditor / Viewer

Responsibilities

- Monitor inventory records
- Review audit logs

Permissions

- Read-only access

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4. Functional Requirements

4.1 Authentication & User Management

Features

- User login/logout
- Password reset
- Session management
- Role-based access control (RBAC)
- Multi-factor authentication (optional)

Functional Requirements

- Users shall authenticate using username/email and password.
- The system shall support role-based permissions.
- The system shall maintain audit logs for user activities.
- The system shall automatically expire inactive sessions.

4.2 Product Management

Features

- Product catalog management
- Product categorization
- SKU generation
- Barcode support

Functional Requirements

- Users shall create, update, and deactivate products.
- Each product shall have:
    - Product ID
    - SKU
    - Name
    - Description
    - Category
    - Unit of measure
    - Supplier information
    - Reorder level
- The system shall prevent duplicate SKUs.
- Products may support barcode/QR code association.

4.3 Inventory Management

Features

- Real-time stock tracking
- Multi-location inventory
- Stock movement history

Functional Requirements

- The system shall maintain current stock quantity per product.
- The system shall track stock by warehouse/location.
- The system shall record:
    - Stock additions
    - Stock removals
    - Transfers
    - Damages
    - Returns
- Inventory updates shall occur in real time.
- Negative inventory shall be restricted unless explicitly allowed.

4.4 Stock Inward Management

Features

- Goods receiving
- Purchase receiving
- Supplier delivery tracking

Functional Requirements

- Users shall record incoming inventory.
- The system shall validate purchase order quantities.
- Received inventory shall automatically update stock levels.
- Partial receiving shall be supported.
- Damaged goods shall be recorded separately.

4.5 Stock Outward Management

Features

- Dispatch processing
- Sales deduction
- Shipment tracking

Functional Requirements

- Users shall record outgoing inventory.
- Stock shall reduce automatically upon dispatch.
- The system shall prevent dispatch beyond available stock.
- Return transactions shall restore inventory.

4.6 Stock Adjustment Management

Features

- Manual adjustments
- Damage/loss recording
- Correction workflows

Functional Requirements

- Authorized users shall adjust stock quantities.
- Adjustment reasons shall be mandatory.
- All adjustments shall be audit logged.
- Approval workflow may be required for large adjustments.

4.7 Warehouse Management

Features

- Multi-warehouse support
- Bin/shelf tracking
- Internal transfers

Functional Requirements

- The system shall support multiple warehouses.
- Products may be assigned storage locations.

- Users shall transfer stock between warehouses.
- Transfer history shall be maintained.

4.8 Supplier Management

Features

- Supplier database
- Supplier-product mapping
- Purchase history

Functional Requirements

- Users shall create and manage suppliers.
- Supplier details shall include:
    - Contact information
    - Address
    - Tax details
- The system shall track supplier transactions.

4.9 Purchase Order Management

Features

- Purchase order creation
- Approval workflow
- PO tracking

Functional Requirements

- Users shall create purchase orders.
- Purchase orders may require approval.
- The system shall track:
    - Ordered quantity
    - Received quantity

    - Pending quantity
- Purchase order statuses:
    - Draft
    - Approved
    - Partially Received
    - Completed
    - Cancelled

4.10 Reporting & Analytics

Features

- Operational reports
- Inventory analytics
- Export functionality

Functional Requirements

- The system shall generate:
    - Stock summary report
    - Inventory valuation report
    - Low stock report
    - Transaction history report
    - Expiry report
- Reports shall support:
    - Filtering
    - Sorting
    - Export to Excel/PDF/CSV

4.11 Alerts & Notifications

Features

- Low stock alerts
- Expiry alerts

- Overstock alerts

Functional Requirements

- The system shall notify users when stock falls below reorder levels.
- Notifications may be sent via:
    - Email
    - SMS
    - In-app alerts
- Expiry alerts shall support configurable thresholds.

4.12 Audit & Logging

Features

- Activity tracking
- Inventory audit logs

Functional Requirements

- All inventory changes shall be logged.
- Logs shall include:
    - User
    - Timestamp
    - Action performed
    - Previous value
    - New value
- Audit logs shall be immutable.

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5. Key Workflows

5.1 Product Creation Workflow

1. User logs in

2. Navigate to Product Management
3. Enter product details
4. Validate SKU uniqueness
5. Save product
6. Product becomes available in inventory

5.2 Stock Receiving Workflow

1. Create/Select Purchase Order
2. Receive goods
3. Verify quantities
4. Record damaged items
5. Update inventory
6. Generate receiving receipt

5.3 Stock Dispatch Workflow

1. Create dispatch request
2. Validate stock availability
3. Approve dispatch
4. Deduct inventory
5. Generate dispatch document

5.4 Stock Transfer Workflow

1. Select source warehouse
2. Select destination warehouse
3. Enter transfer quantity
4. Approve transfer
5. Update inventory in both locations

5.5 Stock Adjustment Workflow

1. Initiate adjustment request
2. Enter reason
3. Approve adjustment
4. Update stock quantity
5. Record audit trail

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6. Non-Functional Requirements (NFRs)

6.1 Performance

- System response time shall be less than 3 seconds for standard operations.
- Inventory updates shall occur in near real time.
- System shall support concurrent users.

6.2 Scalability

- System shall support horizontal scaling.
- Architecture shall support increasing product and transaction volumes.

6.3 Availability

- System uptime target: 99.9%
- Backup and recovery mechanisms shall be implemented.

6.4 Security

- Passwords shall be encrypted.
- HTTPS shall be enforced.
- RBAC shall restrict unauthorized access.
- Sensitive actions shall require authorization.
- Audit logs shall be retained securely.

6.5 Reliability

- Transactions shall be ACID compliant.
- Inventory updates shall maintain consistency.
- Data corruption prevention mechanisms shall exist.

6.6 Maintainability

- Modular architecture shall be used.
- APIs shall follow standard documentation practices.
- Logging and monitoring shall be implemented.

6.7 Usability

- UI shall be responsive and user friendly.
- Mobile/tablet support should be available.
- Minimal training should be required.

6.8 Compatibility

- System shall support modern browsers.
- APIs shall support third-party integrations.

6.9 Compliance

- System shall comply with organizational audit requirements.
- Data retention policies shall be configurable.

6.10 Backup & Disaster Recovery

- Daily automated backups
- Point-in-time recovery support
- Disaster recovery procedures documented

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7. Suggested Future Enhancements

- Barcode scanner integration
- QR code inventory tracking
- RFID support
- Mobile inventory app
- AI-based demand forecasting
- ERP/accounting integration
- Multi-tenant architecture
- Offline warehouse mode